A waist at the binary collision: two-letter syzygy orbits of the strong-force three-body problem

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Abstract

Consider the planar three-body problem with the strong-force potential $U = \sum m_i m_j / r_{ij}^2$ at zero energy and zero angular momentum. Modulo symmetries its solutions are the geodesics of Montgomery's Jacobi-Maupertuis metric on the shape sphere minus the three binary collisions, a pair of pants whose ends are asymptotic to cylinders. For equal masses the two-letter syzygy classes $\dots ijij \dots$, the loops around a single binary collision, carry no solution. We show that the Gaussian curvature at a binary collision is, to leading order, $K = \kappa_2 \rho^2$ with

$$\kappa_2 = -\frac{(m_i + m_j)\big[m_k^2 (m_i + m_j)^3 - m_i^2 m_j^2 M\big]}{4M\, m_i^4 m_j^4}, \qquad M = m_1 + m_2 + m_3,$$

the leading coefficient of Montgomery's curvature formula at the collision. When $\kappa_2 > 0$, that is $m_k^2 (m_i + m_j)^3 < m_i^2 m_j^2 M$, the end has a *waist*, a loop shorter than the asymptotic cylinder, and the class (ij) is realised by a collision-free solution, periodic up to rotation, whose action lies below the cylinder's circumference. At most one end can have a waist, that of the two heaviest bodies; for masses $(1, 1, m)$ the condition is $m < m^* = (1 + \sqrt{65})/16 \approx 0.5664$. We compute these orbits at nine mass triples (all hyperbolic) and show that the class is born from the cylinder at m^*: the action of the minimiser satisfies $2\pi - S \geq 35.12\,(m^* - m)^2 (1 + o(1))$, and the computed actions approach this rate. We give a short Gauss-Bonnet proof that no two-letter orbit exists for equal masses, and, for masses without a waist at any end, reduce the non-existence to three conditions on families of curves; we prove the first by interval arithmetic at five mass triples and on the family $(1, 1, m)$, $0.567 \leq m \leq 1.07$, and check the other two numerically at five triples. A map of the whole mass triangle agrees with the waist criterion at all 741 grid points. The orbits, the search, the threshold and the converse checks come with independent verification scripts.

1 Introduction

Montgomery [15] showed that for the planar three-body problem with an attractive $1/r^2$ potential, zero energy and zero angular momentum, the solutions are, after reduction by rotations, translations and scaling, the geodesics of a complete Riemannian metric on the shape sphere with the three binary collision points removed: a pair of pants whose three ends are asymptotic to cylinders. For equal masses this metric has negative curvature except at the two Lagrange points; as a consequence every free homotopy class of loops contains at most one closed geodesic,

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the syzygy sequences give a complete symbolic dynamics, and the classes coded by two-letter sequences $\dots ijij \dots$, the loops around a single binary collision, are not realised by any bounded solution [15, Thm. 3]. Montgomery also gave the curvature for arbitrary masses in closed form [15, (7.12)-(7.13)] and showed that for most unequal masses it takes both signs near the Lagrange points [15, Thm. 6]. (We use the numbering of the arXiv version of [15].)

The two-letter classes are the untied classes in the sense of Montgomery [14] (after Gordon [6, 7]). In the realisation results for the strong force they are the exceptions: every other class is realised, and the two-letter classes are stated not to be, without a restriction on the masses [15, §2.1], [12, §1]. Near each end the curvature is negative whenever $\kappa_2 < 0$ (Lemma 2), in particular for equal masses, and the end then approaches its cylinder monotonically [15, remark after (3.18)]; the proof of non-existence for equal masses [15, §6.3] uses this monotone approach together with the isosceles reflections, which are isometries for equal masses. For the Newtonian problem, [12, open question 6] names planetary and earth-moon-sun type configurations as realising the two-letter classes. Problem 4 of the list [1] (Montgomery's) asks which syzygy sequences are realised in the Newtonian problem; its comment records, for the strong force with equal masses, the correspondence of the bounded collision-free orbits with the sequences in which no letter repeats and no pair of letters repeats forever [15]. With a waist the periodic sequence $\dots ijij \dots$ is realised (Theorem 6), so that description fails whenever an end has a waist; whether it holds when no end has one is open (Section 6.2).

The present note studies the two-letter classes for unequal masses. The curvature tends to zero at each end; the question is the sign of the next term. We compute its leading coefficient κ_2 in closed form (Lemma 2). Its sign is the limit at the collision of Montgomery's sign function [15, (7.13)], a fact the supplement checks symbolically, and it has a direct geometric meaning: $\kappa_2 > 0$, i.e. $m_k^2(m_i + m_j)^3 < m_i^2 m_j^2 M$, exactly when the end has a waist, a loop shorter than its asymptotic cylinder, so that the approach of the end to its cylinder is not monotone. In that case the class (ij) is realised (Theorem 6). For equal masses $\kappa_2 = -5/6$, in accordance with [15]. We describe the orbits numerically, show how they are born from the cylinder as the masses cross the threshold, and treat the converse: without a waist, the non-existence reduces to three conditions on families of curves; we prove the first by interval arithmetic at five mass triples and on the family $(1, 1, m)$, $0.567 \le m \le 1.07$, and check the other two numerically at five triples. Numerical studies of the zero-angular-momentum strong-force problem, mostly for equal masses, include [16, 4]; the curvature of the same metric, and the cylinder at a collision for arbitrary masses, are studied in [10, §3.3.1]. Computer-assisted proofs now settle questions about periodic three-body orbits [3]. For the Newtonian problem with unequal masses large catalogues of periodic orbits have been computed [11]; the strong-force problem offers a setting in which part of such a picture can be proved.

2 Setting

Let $q_1, q_2, q_3 \in \mathbb{R}^2$ with masses $m_i > 0$, centre of mass at the origin, $U = \sum_{i<j} m_i m_j / r_{ij}^2$, energy $E = 0$ and angular momentum $L = 0$. Since U is homogeneous of degree -2, the Lagrange-Jacobi identity gives $\ddot{I} = 4E = 0$ for $I = \sum m_i |q_i|^2$, so a bounded solution has constant I, which we normalise to 1.

Shape sphere. With mass-weighted Jacobi vectors $z_1 = \sqrt{\mu_{12}}\,(q_2 - q_1)$ and $z_2 = \sqrt{\mu_3}\,\big(q_3 - \frac{m_1 q_1 + m_2 q_2}{m_1 + m_2}\big)$, where $\mu_{12} = m_1 m_2/(m_1 + m_2)$ and $\mu_3 = m_3(m_1 + m_2)/M$, we have $I = |z_1|^2 + |z_2|^2$,

and the Hopf map $w = (|z_1|^2 - |z_2|^2,\, 2z_1\bar{z}_2) \in S^2$ identifies shapes (triangles up to rotation and scale) with points of the unit sphere. The kinetic metric descends to the round sphere of radius $1/2$, $g_0 = \frac{1}{4} g_{S^2}$. Each r_{ij}^2 is an affine function of w; for instance $r_{12}^2 = (1 + w_1)/(2\mu_{12})$. The equator $w_3 = 0$ consists of the collinear shapes; it is cut by the three collision points C_{12}, C_{13}, C_{23} into three arcs, arc k being the shapes with body k in the middle.

The metric. Solutions with $E = 0$, $L = 0$, $I \equiv 1$ project to geodesics of $g = 2U\, g_0$ on $P = S^2 \setminus \{C_{12}, C_{13}, C_{23}\}$, and conversely [15]. The action $\int 2T\, dt$ of a solution equals the g-length of its projection; g is twice the metric of [15, (3.9a)], which is why lengths are actions, and our ρ below is the unit-sphere distance. Near C_{ij} the metric is asymptotic to a Euclidean cylinder of circumference $c_{ij} = 2\pi\sqrt{2\mu_{ij} m_i m_j}$, $\mu_{ij} = m_i m_j/(m_i + m_j)$ [15, §3]; Lemma 1 below gives the precise form. A closed geodesic gives a solution that is periodic up to a rotation of the whole configuration.

Syzygies. The syzygy sequence of a curve lists which body is in the middle at each crossing of the equator. Cancelling repeated letters (stutters) reduces it; free homotopy classes of loops in P correspond to reduced periodic sequences up to cyclic shift, the reduced sequence of a closed curve being that of its class [15]. The two-letter class (ij), the loop around C_{ij}, has the sequence $\ldots ijij \ldots$.

3 The ends and the curvature at a binary collision

Fix the collision C_{12}; the other two are treated alike. Let ρ be the unit-sphere distance from C_{12} and χ the angle about it. At $I = 1$, $r_{12}^2 = \sin^2(\rho/2)/\mu_{12}$, so

$$U = \frac{\mu_{12} m_1 m_2}{\sin^2(\rho/2)} + V,$$

where V, the potential of the pairs 13 and 23, is smooth near C_{12} with value $V_0 = m_3^2(m_1 + m_2)^2/M$ there: at the collision configuration body 3 is at distance D from the coincident pair with $\mu_3 D^2 = 1$, so $V_0 = (m_1 + m_2)m_3/D^2$. We write $R^2 = 2\mu_{12} m_1 m_2$, so that $c_{12} = 2\pi R$.

Lemma 1 (the end in conformal coordinates)**.** *With $u = -\log\tan(\rho/2)$,*

$$g = R^2 e^{2\psi}\,(du^2 + d\chi^2), \qquad e^{2\psi} = \cos^2(\rho/2)\,(1 + \varepsilon), \qquad \varepsilon = \frac{\sin^2(\rho/2)\, V}{\mu_{12} m_1 m_2}.$$

The function ψ is smooth in $\zeta = \tan(\rho/2)e^{i\chi}$ near $\zeta = 0$, $\psi = |\zeta|^2 h(\zeta)$ with h smooth and $h(0) = \frac{1}{2}\big(V_0/(\mu_{12} m_1 m_2) - 1\big)$. In particular ψ and all its derivatives in (u, χ) are $O(e^{-2u})$, uniformly in χ.

Proof. $g_0 = \frac{1}{4}(d\rho^2 + \sin^2\rho\, d\chi^2) = \frac{1}{4}\sin^2\rho\,(du^2 + d\chi^2)$ since $du = -d\rho/\sin\rho$, and $\frac{1}{2}U\sin^2\rho = 2\mu_{12} m_1 m_2 \cos^2(\rho/2)(1 + \varepsilon)$. The coordinate ζ is the stereographic coordinate centred at C_{12}, $\sin^2(\rho/2) = |\zeta|^2/(1 + |\zeta|^2)$ and $\cos^2(\rho/2) = 1/(1 + |\zeta|^2)$, so $2\psi = -\log(1 + |\zeta|^2) + \log(1 + \varepsilon)$ is $|\zeta|^2$ times a smooth function with the stated value at 0. Since $|\zeta| = e^{-u}$, $\partial_u = -(\xi\partial_\xi + \eta\partial_\eta)$ and $\partial_\chi = \xi\partial_\eta - \eta\partial_\xi$ ($\zeta = \xi + i\eta$), and both operators map $|\zeta|^2 \cdot (\text{smooth})$ into itself. □

Thus each end is a half-cylinder of circumference c_{12} up to an exponentially small conformal factor; in particular g is complete. Two consequences are used below. A loop in $\{u \geq u_0\}$ winding once around the end has length at least $c_{12}\, e^{-Ce^{-2u_0}}$, and an arc joining the levels $u_1 < u_2$ has length at least $R\, e^{-Ce^{-2u_1}}(u_2 - u_1)$, where $|\psi| \leq Ce^{-2u}$ as in the lemma. The loops $u = \Lambda$ have total geodesic curvature $\oint \partial_u \psi \, d\chi = O(e^{-2\Lambda})$. Finally, the reflection $w_3 \mapsto -w_3$ is an isometry of g for all masses (each r_{ij}^2 depends on w_1, w_2 only), so the three arcs of the equator are geodesics.

Lemma 2. *The Gaussian curvature of g satisfies*

$$K = \kappa_2\, \rho^2 + O(\rho^3),$$
$$\kappa_2 = \frac{1}{4\mu_{12} m_1 m_2}\Big(1 - \frac{V_0}{\mu_{12} m_1 m_2}\Big) = -\frac{(m_1+m_2)\big[m_3^2(m_1+m_2)^3 - m_1^2 m_2^2 M\big]}{4M\, m_1^4 m_2^4},$$

independently of the direction.

Proof. By Lemma 1, $K = -R^{-2}e^{-2\psi}(\psi_{uu} + \psi_{\chi\chi})$ and $\psi = h(0)e^{-2u} + O(e^{-3u})$ with derivatives, so $K = -4h(0)R^{-2}e^{-2u} + O(e^{-3u})$. Since $e^{-2u} = \tan^2(\rho/2) = \rho^2/4 + O(\rho^4)$, this is $K = -h(0)\rho^2/R^2 + O(\rho^3)$, which is the first expression; the second follows from $\mu_{12} m_1 m_2 = m_1^2 m_2^2/(m_1+m_2)$ and the value of V_0. The direction does not enter because the leading term of V at C_{12} is the constant V_0: the third body is equidistant from the coincident pair. □

Remark 3. A second derivation works on the unit sphere. Writing $g = e^{2\phi} g_{S^2}$ with $e^{2\phi} = U/2$, one has $K = e^{-2\phi}(1 - \Delta_S \phi) = (4 - 2\Delta_S \log U)/(2U)$, and $\Delta_S \log(1 - \cos\rho) = -1$ exactly; expanding $\log U$ gives the same κ_2. The $O(\rho^3)$ term comes from $2\sin\rho\, \partial_\rho V$ and depends on the direction. At equal masses $\kappa_2 = -5/6$. Exact rational expansions at ten mass triples, and numerical checks at the five triples of Section 6.2 and its control, agree with the formula (supplement).

Corollary 4 (waist criterion)**.** *The conditions (a) $m_3^2(m_1+m_2)^3 < m_1^2 m_2^2 M$, (b) $V_0 < \mu_{12} m_1 m_2$, (c) $\kappa_2 > 0$ are equivalent. The round loops $\rho = \text{const}$ about C_{12} have length*

$$\ell(\rho) = c_{12}\cos(\rho/2)\Big\langle \sqrt{1+\varepsilon}\Big\rangle_\chi = c_{12}\Big[1 + \frac{\rho^2}{8}\Big(\frac{V_0}{\mu_{12} m_1 m_2} - 1\Big)\Big] + O(\rho^3),$$

with $\langle\cdot\rangle_\chi$ the mean over χ. Hence if $\kappa_2 > 0$ they are shorter than the cuff for small ρ (the end has a waist*), and if $\kappa_2 < 0$ they are longer. For masses $(1,1,m)$ condition (a) reads $8m^2 - m - 2 < 0$, i.e. $m < m^* = (1+\sqrt{65})/16$.*

Proof. The equivalence of (a), (b), (c) is algebra. By Lemma 1 the length is $\oint R e^{\psi} d\chi$, and $\varepsilon = \rho^2 V_0/(4\mu_{12} m_1 m_2) + O(\rho^3)$; the term of V linear in the direction contributes only at order ρ^3. Numerically, the configuration-space loop lengths agree with this expansion at fourteen mass triples, including $m = 0.566$ and 0.567 on the two sides of m^* (supplement). □

At $\kappa_2 = 0$ ($m = m^*$ for masses $(1,1,m)$) the leading order decides nothing; we leave that single case aside.

Proposition 5. *At most one end has a waist, and it is the binary of the two heaviest bodies.*

Proof. Waists at the ends 12 and 13 would give $m_3^2(m_1+m_2)^3 < m_1^2 m_2^2 M$ and $m_2^2(m_1+m_3)^3 < m_1^2 m_3^2 M$, hence $(m_1+m_2)^3(m_1+m_3)^3 < m_1^4 M^2$. But $(m_1+m_2)(m_1+m_3) = m_1 M + m_2 m_3 > m_1 M$, so $m_1^3 M^3 < m_1^4 M^2$, i.e. $M < m_1$, impossible. If the end ij has a waist and $m_k \geq m_i$, write $s = m_i + m_j$; then $m_k^2 s^3 < m_i^2 m_j^2 (s+m_k)$, and $(m_i/m_k)^2(s+m_k) \leq s + m_i$ for $m_i \leq m_k$, so $s^3 < m_j^2(s+m_i)$; but $s^3 - m_j^2(s+m_i) = m_i(m_i^2 + 3m_i m_j + m_j^2) > 0$. So $m_k < m_i$, and likewise $m_k < m_j$. □

4 Existence

Theorem 6. *If the end ij has a waist, the two-letter class (ij) contains a closed geodesic of g of length $L^* = \inf\{\text{length of loops in } (ij)\} < c_{ij}$. Hence there is a collision-free solution with $E = 0$, $L = 0$, $I = 1$, periodic up to a rotation, realising the class (ij), with action $L^* < c_{ij}$.*

Proof. By the waist, $L^* < c_{ij}$. Take loops γ_n in the class, parametrised proportionally to arc length, with lengths $L_n \downarrow L^*$ and $L_n < L_1 < c_{ij}$. Let u_{ij}, u_{ik}, u_{jk} be the coordinates of Lemma 1 at the three ends. No loop of the class lies in an end $\{u_{ik} \geq u_0\}$: such an end is an annulus whose loops are powers of the class (ik), and the three boundary classes of the pair of pants are distinct and primitive. So each γ_n meets $\{u_{ik} \leq u_0\}$, and by the bound on arcs it reaches at most depth $u_0 + O(L_1)$ in that end; likewise in the end jk. In the end ij, a loop of the class in $\{u_{ij} \geq \Lambda\}$ winds once and has length at least $c_{ij} e^{-Ce^{-2\Lambda}} > L_1$ for Λ large, so each γ_n meets $\{u_{ij} \leq \Lambda\}$ and again goes at most a bounded depth further. Hence all γ_n lie in one compact set $Q \subset P$. They are equi-Lipschitz, so by Arzelà-Ascoli a subsequence converges uniformly to a loop $\gamma \subset Q$ of length at most L^*; once γ_n is uniformly closer to γ than the injectivity radius of Q, the two are freely homotopic, so γ is in the class and has length L^*. A shortest loop in a non-trivial free homotopy class is locally length minimising, hence a smooth closed geodesic.

For the solution: with $r = \sqrt{I}$ the kinetic metric on the configurations with $L = 0$, modulo rotations, is $dr^2 + r^2 g_0$, and the Jacobi-Maupertuis metric at $E = 0$ is $2(U/r^2)(dr^2 + r^2 g_0) = 2U\,(d\log r)^2 + g$ with U the shape potential. It does not depend on $\log r$, so at zero conjugate momentum ($\dot{I} = 0$) its geodesics are the curves with r constant whose shape is a g-geodesic. After the time change $dt = ds/(2U)$ these are solutions with $E = 0$ and $I = 1$, and the horizontal lift through the Hopf map gives $L = 0$; the lift of a closed curve closes up to a rotation. The solution is collision-free because γ avoids the three binary collisions and $I = 1$ excludes the triple collision. Its action $\int 2T\,dt$ is the g-length L^*. □

5 The orbits

We computed the orbits by a one-parameter reversible search: a syzygy with body 1 between bodies 2 and 3, all velocities perpendicular to the line, constrained by $P = 0$, $L = 0$, $E = 0$, with the position of body 1 on the segment as the parameter; a root of the along-line shape velocity at the next syzygy (body 2 in the middle) closes the orbit by reversibility after twice that time. Table 1 lists the orbits found; for $m = 0.57, 0.58, 0.59, 0.60, 0.65, 0.75$ (masses $(1, 1, m)$) and for $(1, 0.5, 0.45)$, $(1, 0.2, 0.19)$ and the controls $(1, 0.3, 0.5)$, $(0.5, 0.3, 1)$ (no waist at the end 12; both have one at the end 13 of their two heaviest bodies, as Proposition 5 requires) no root exists, consistent with Corollary 4 and Proposition 5; all nineteen cases are rerun in the supplement. The search sees only orbits with this reversing symmetry; orbits without it are the subject of

Section 6.2. The class (12) is primitive, so each orbit runs once around C_{12} and its syzygy word per period is 12. Times are for $I = 1$; the action is scale invariant.

masses	action S	cuff c_{12}	period	trace	rotation
$(1, 1, 0.40)$	5.6342554918	2π	1.71728	53.50	$+146.0°$
$(1, 1, 0.50)$	6.1597496557	2π	1.40783	8.102	$-116.9°$
$(1, 1, 0.52)$	6.2197385980	2π	1.18199	4.988	$-89.0°$
$(1, 1, 0.53)$	6.2429769247	2π	1.02557	3.872	$-73.3°$
$(1, 1, 0.54)$	6.2613331277	2π	0.82939	3.017	$-56.1°$
$(1, 1, 0.55)$	6.2744359523	2π	0.58067	2.413	$-37.0°$
$(1, 1, 0.56)$	6.2817961704	2π	0.25915	2.068	$-15.5°$
$(1, 0.5, 0.30)$	3.4385697805	$2\pi/\sqrt{3}$	2.88203	22.01	$-171.1°$
$(1, 0.2, 0.15)$	1.5489069692	$2\pi/\sqrt{15}$	6.38024	19.94	$-169.0°$

Table 1: Closed orbits in the class (12); syzygy word 12 per period, shape closure $7 \cdot 10^{-12}$ to $2 \cdot 10^{-9}$. Trace: of the linearised return map of the closed geodesic; rotation: of the whole configuration per period, counterclockwise positive, in $(-180°, 180°]$ (so $+146.0°$ is the same as $-214.0°$).

Stability. The normal Jacobi equation $J'' + KJ = 0$ along each closed geodesic gives the 2×2 linearised return map of the geodesic flow on the shape sphere, i.e. of the reduced flow; the full relative periodic orbit has in addition the trivial multipliers of the symmetries, the integrals and the dilation. The trace (Table 1) exceeds 2 for all nine orbits, so they are hyperbolic as closed geodesics of g, with multiplier $\lambda = (\mathrm{tr} + \sqrt{\mathrm{tr}^2 - 4})/2$ from 53.5 at $m = 0.40$ down to 1.30 at $m = 0.56$, tending to 1 as $m \to m^*$. This is the behaviour of a non-degenerate closed geodesic of index 0 on an orientable surface [9], although K is positive on part of every orbit (its maximum on the orbit is 0.003 at $m = 0.56$ and 0.92 at $(1, 0.2, 0.15)$). The rotation per period is the geometric phase of the shape curve [13]: half the solid angle it encloses on the unit shape sphere, which the supplement checks against the rotation to $0.05°$.

Birth from the cylinder. For masses $(1, 1, m)$ the numerical threshold lies in $(0.56, 0.57)$, containing $m^* = 0.56639$. The round loops have length $\ell(\rho) = c_{12}\big[1 - A_2\rho^2 + A_4\rho^4 + O(\rho^6)\big]$, the odd orders averaging out, with $A_2 = (1 - v)/8 = \mu_{12} m_1 m_2 \, \kappa_2/2$, $v = V_0/(\mu_{12} m_1 m_2)$, and

$$A_4 = \frac{(1 - v)(1 + 3v)}{384} + \frac{m_3^3 (m_1 + m_2)^3}{32 M^2 m_1^2 m_2^2}.$$

Indeed $\ell = c_{12} \cos(\rho/2) \langle \sqrt{1 + \varepsilon} \rangle_\chi$ (Corollary 4), the mean of V over the circle is $V_0 + \frac{\rho^2}{4} \Delta_S V(C_{12}) + O(\rho^4)$, and since r_{13}^2 and r_{23}^2 are affine in w one finds $\Delta_S V(C_{12}) = V_0 + m_3 \mu_3^2$; expanding to order ρ^4 gives the formula. On the threshold curve $v = 1$, so

$$A_4 = \frac{m_3^3 (m_1 + m_2)^3}{32 M^2 m_1^2 m_2^2} > 0$$

for all masses: the second coefficient is positive along the whole boundary of the waist region. For masses $(1, 1, m)$, $A_4(m^*) = m^{*3}/(4(2 + m^*)^2) = 0.0068967$. Minimising over ρ, the shortest

round loop has

$$c_{12} - \min_{\rho} \ell = \frac{c_{12}A_2^2}{4A_4}\big(1+o(1)\big) = C\,(m^*-m)^2\big(1+o(1)\big),$$
$$C = \frac{c_{12}A_2'(m^*)^2}{4A_4(m^*)} = \frac{\pi(4+m^*)^2}{32\,m^{*5}} = 35.1205,$$

at $\rho^2 \approx A_2/(2A_4) \approx 28.5\,(m^*-m)$. The minimiser of Theorem 6 has length $L^* \le \min_\rho \ell$, so $2\pi - L^* \ge C\,(m^*-m)^2(1+o(1))$; that the computed symmetric orbits are the minimisers we have not proved, but their actions satisfy $S < \min_\rho \ell$ at all nine masses (by 10^{-6} at $m = 0.56$, supplement). Numerically the ratios $(2\pi - S)/(m^*-m)^2$ from Table 1 are 28.00, 29.48, 30.36, 31.37, 32.57, 34.01 for $m = 0.50$, 0.52 to 0.56, and those of the shortest round loops are 31.9, 33.4, 34.2, 34.7, 34.9 at $m^* - m = 0.02$, 0.01, 0.005, 0.0025, 0.00125: both rise towards C (Figure 3). As $m \to m^*$ the orbit shrinks into the end (Figure 1), with the period and the binary separation tending to zero. The sign $A_4 > 0$ is what makes the shortest round loop leave the cuff continuously at m^*; the search finds no orbit of the class above m^* (no fold).

The motion. At $L = 0$ the binary axis and the line from the binary's centre of mass to body 3 turn in opposite senses (Figure 2): over one period by $+344.5°$ and $-15.5°$ at $m = 0.56$, and by $+146.0°$ and $-214.0°$ at $m = 0.40$. The reason is the angular momentum: a circular binary of the $1/r^2$ problem with zero energy has angular momentum exactly $\sqrt{2\mu_{12}m_1m_2} = R$, whatever its size, so at $L = 0$ the outer motion carries $-R$, and near the threshold the line to body 3 turns back by about R times the period ($14.8°$ predicted, $15.5°$ observed at $m = 0.56$). Near the threshold the motion is hierarchical: a tight binary spins about once per period while the third body swings slightly back (binary separation 0.29, distance of body 3 from the binary's centre 1.48 at $m = 0.56$). Away from it the two distances are comparable (1.05 to 1.13 against 1.05 to 1.16 at $m = 0.40$). The cylinder end corresponds to the circular binary motions of the $1/r^2$ two-body problem, which have zero energy at every radius; the third body's potential V_0 competes with the factor $\cos(\rho/2)$ by which the sphere $I = 1$ shrinks the loops, and the waist appears when $V_0 < \mu_{12}m_1m_2$. In terms of angular momentum: a pair with reduced mass μ in the potential k/r^2 at zero energy reaches collision unless its angular momentum is at least $\sqrt{2\mu k}$. The binary on the cylinder sits exactly at its critical value R, and the outer pair, body 3 against the coincident binary (μ_3, $k = m_3(m_1+m_2)$), has critical value $\sqrt{2\mu_3 m_3(m_1+m_2)} = \sqrt{2V_0}$. At $L = 0$ the outer motion carries $-R$, and the criterion $V_0 < \mu_{12}m_1m_2$ reads $\sqrt{2V_0} < R$: the waist exists exactly when the angular momentum the binary forces on the outer motion exceeds the outer pair's own critical value. In Euler's two-centre problem the corresponding orbits are the planetary orbits of [5], with fixed centres; for the Newtonian three-body problem compare [12, open question 6].

6 The converse

6.1 Equal masses

Lemma 7. *If γ is a simple closed geodesic in the class (ij), the region A between γ and the end ij has $\int_A K\,dA = 0$.*

Proof. A simple closed curve on the sphere minus three points separates the points as $0|3$ or $1|2$, so it is null-homotopic or isotopic to the loops around one puncture; hence γ is isotopic to

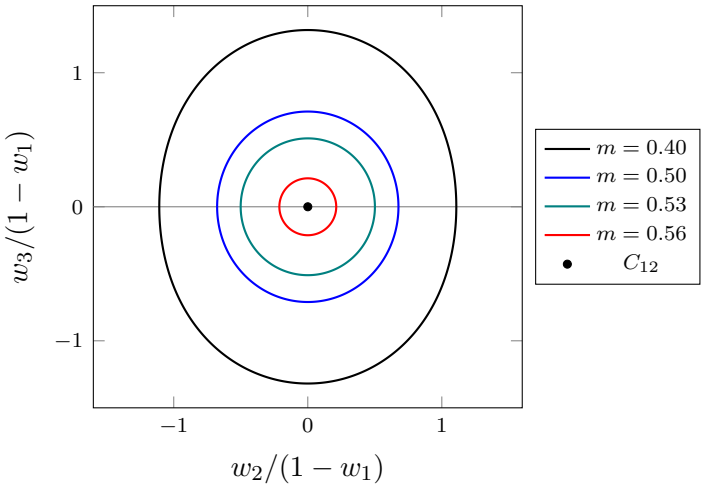


Figure 1: The orbits of masses $(1, 1, m)$ on the shape sphere, in the stereographic chart centred at C_{12}; the horizontal axis is the equator (collinear shapes). As $m \to m^*$ the orbit contracts into the end.

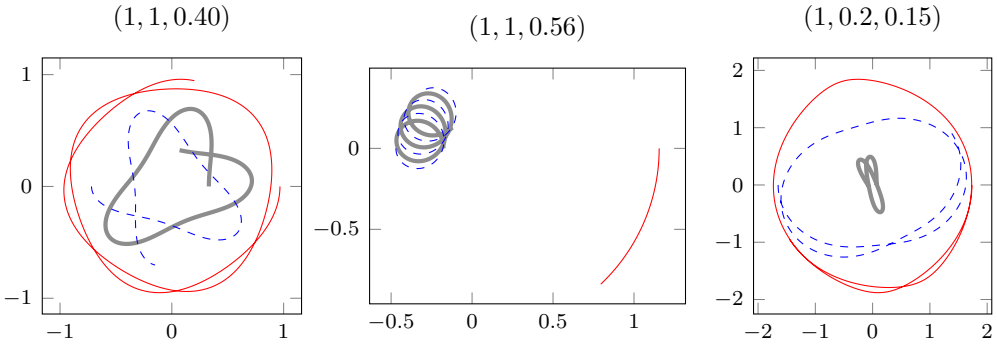


Figure 2: Three periods in the plane, centre of mass at the origin: body 1 grey (wide), body 2 blue (dashed), body 3 red; at $(1, 1, 0.56)$ the paths of bodies 1 and 2 nearly coincide. The configuration returns rotated by the angle of Table 1.

the loops $u_{ij} = \text{const}$ and A is an annulus running into the end. Cut it at $u_{ij} = \Lambda$ and apply Gauss-Bonnet to the compact annulus (Euler characteristic 0): γ has zero geodesic curvature and the loop $u_{ij} = \Lambda$ has total geodesic curvature $O(e^{-2\Lambda})$ by Lemma 1. The integral converges absolutely, as $K\,dA = O(e^{-2u})\,du\,d\chi$ in the end. □

Corollary 8 ([15, Thm. 3])**.** *For equal masses no two-letter class is realised.*

Proof. Here $K \leq 0$, with equality only at the two Lagrange points [15]. Let γ be a closed geodesic in the class (ij), φ the corresponding deck transformation of the universal cover X of (P, g), and $\hat{P} = X/\langle\varphi\rangle$ the annular cover. X is complete, simply connected and has $K \leq 0$, so its geodesics are embedded lines and a lift $\tilde{\gamma}$ of γ is an axis of φ [2, II.6]. The preimage in X of its image $\hat{\gamma} \subset \hat{P}$ is $\bigcup_k \varphi^k\tilde{\gamma} = \tilde{\gamma}$, so $\hat{\gamma}$ is a simple closed geodesic in the core class of $\hat{P}$. The end ij lifts isometrically to $\hat{P}$, its fundamental group being $\langle\varphi\rangle$; $\hat{\gamma}$ cuts $\hat{P}$ into two annuli, one running into this end, and the proof of Lemma 7 applies there word for word: $\int K\,dA = 0$ over

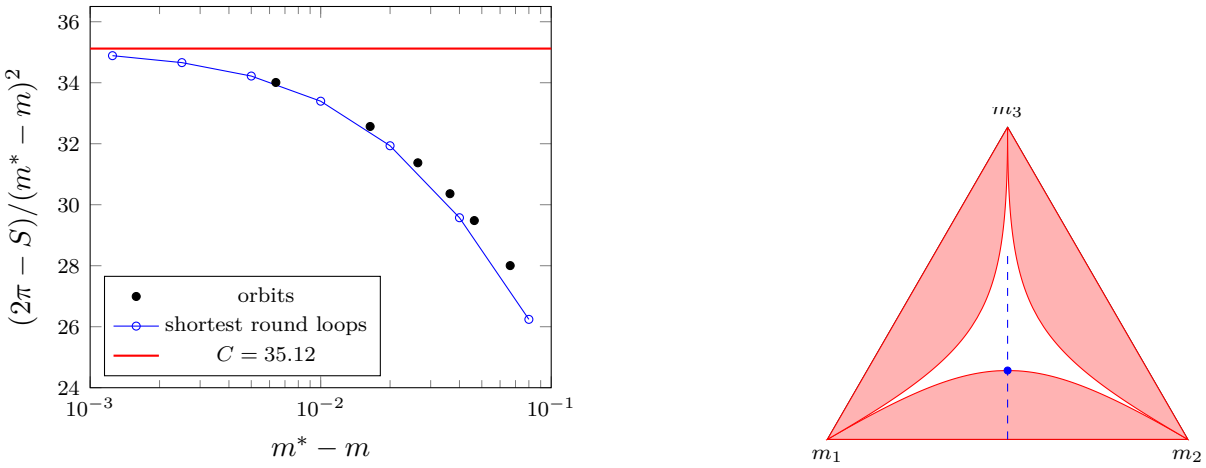


Figure 3: Left: the birth of the orbit from the cylinder, masses $(1, 1, m)$: the ratio $(2\pi - S)/(m^* - m)^2$ for the orbits of Table 1 and for the shortest round loops about C_{12}, rising towards the constant C. Right: the mass triangle $(m_1 + m_2 + m_3 = 1)$; shaded, the masses for which the end of the two bodies on the nearby edge has a waist (Corollary 4); the regions are disjoint (Proposition 5). Dashed, the masses $(1, 1, m)$; dot, $m = m^*$.

a region where $K \le 0$, with $K < 0$ off the discrete preimage of the Lagrange points, which is impossible. □

6.2 Unequal masses without a waist

Throughout this subsection no end has a waist: $\kappa_2 < 0$ at all three collisions, so by Lemma 2 the curvature is negative in a punctured neighbourhood of each collision. (By Proposition 5 a waist can occur at one end only; when it occurs at an end other than 12, the class (12) is not covered by what follows.) We reduce the non-existence of a closed geodesic in the class (12) to three conditions, each a check on one-parameter families of curves.

Lemma 9 (maximum principle)**.** *Let τ be a C^2 function with $\nabla\tau \neq 0$ on an open set Ω, whose level sets have geodesic curvature strictly positive towards increasing τ; equivalently* $\operatorname{Hess}\tau(v, v) < 0$ *for $v \neq 0$ tangent to a level set, the geodesic curvature being $-\operatorname{Hess}\tau(v, v)/|\nabla\tau|$ for unit v. If a geodesic γ is tangent at $\gamma(0) \in \Omega$ to the level set through it, then $\tau(\gamma(t)) < \tau(\gamma(0))$ for all small $t \neq 0$. In particular no geodesic arc in Ω has an interior local minimum of τ, and no geodesic touches a level set from the side of increasing τ.*

Proof. $(\tau \circ \gamma)'(0) = 0$ and $(\tau \circ \gamma)''(0) = \operatorname{Hess}\tau(\gamma'(0), \gamma'(0)) < 0$. □

Conditions. Let σ_0 be a smooth simple loop around C_{12}, bounding a disc $D \ni C_{12}$ that contains no other collision, with geodesic curvature positive towards C_{12} (convex), and disjoint from arc 3. Let $\Omega_0 = P \setminus \overline{D}$.

(I) $K \le 0$ on Ω_0.

(II) There are smooth convex loops $\sigma_0 \supset \sigma_1 \supset \cdots \supset \sigma_N$ around C_{12}, each inside the previous one, and times $T_k > 0$, with the following property. For each $k < N$ the normal map

$\Phi_k(s,t) = \exp(t\,n(s))$, $s \in \sigma_k$, $0 \le t \le T_k$, n the inward unit normal, is injective; along every normal the Jacobi field $G'' = -KG$, $G(0) = 1$, $G'(0) = -k_g(s)$ has $G > 0$ and $G' < 0$ on $[0, T_k]$ (the parallels are immersed and convex); and the parallel $\Phi_k(\cdot, T_k)$ lies inside σ_{k+1}. The last loop σ_N lies in a cap $\rho < \rho_N$ about C_{12} on which the round circles are convex towards C_{12} (condition C: $2\cot\rho + \partial_\rho \log U > 0$).

(III) No stutter: no geodesic arc lies, apart from its two endpoints, in an open hemisphere and has both endpoints on the same arc of the equator.

Near C_{12} condition C holds automatically when $\kappa_2 < 0$: half its left side is the ρ-derivative of the logarithm of the length density of the circle $\rho = \text{const}$, which by Corollary 4 is

$$\tfrac{1}{2}\big(2\cot\rho + \partial_\rho \log U\big) = \frac{\rho}{4}\Big(\frac{V_0}{\mu_{12} m_1 m_2} - 1\Big) + O(\rho^2).$$

Proposition 10. *Under (I), (II), (III) the class* (12) *contains no closed geodesic.*

Proof. Let γ be a closed geodesic in the class (12).

Syzygies. The sequence of γ is not empty, the class being non-trivial. A stutter in it is an arc as excluded in (III), so the sequence is reduced; a reduced sequence determines the free homotopy class and conversely [15], so it is $1212\ldots$ and γ does not cross arc 3. It does not touch it either, since arc 3 is a geodesic.

γ lies in $\overline{D}$. The closure of Ω_0 in S^2 is a disc containing C_{13}, C_{23} and arc 3. A geodesic tangent to σ_0 stays locally outside D (Lemma 9 for the distance to σ_0), so γ crosses σ_0 transversally wherever it enters D, and by compactness meets σ_0 finitely often. Suppose first that γ meets D and Ω_0, and let β be an arc of γ in Ω_0 with endpoints on σ_0. If β is not simple, its first return to one of its own points closes a geodesic monogon in $\overline{\Omega_0}$, excluded as below. If β is simple, it cuts the disc $\overline{\Omega_0}$ in two; arc 3, which β avoids, lies in one of them, and the other, Δ, is a disc without punctures bounded by β and an arc of σ_0. Gauss-Bonnet on Δ, with exterior angles $\theta_1, \theta_2 \in (0,\pi)$ and the arc of σ_0 curving away from Δ, gives $\int_\Delta K = 2\pi - \theta_1 - \theta_2 + \int_{\sigma_0 \cap \partial\Delta} k_g > 0$, against (I). Suppose next that $\gamma \subset \overline{\Omega_0}$; it may touch σ_0 from outside at finitely many points. Then γ lies in the annulus $\overline{\Omega_0}\backslash$arc 3, in its core class. If γ is simple it bounds with σ_0 an annulus A', or, if it touches σ_0, a chain of discs with two cusps each (exterior angles π); in both cases Gauss-Bonnet gives $\int K = \oint_{\sigma_0} k_g > 0$ (k_g towards C_{12}), against (I). If not, γ contains a geodesic monogon β with interior angle $\varphi \in (0,\pi)$ (the same evaluations hold if its corner lies on σ_0): if β bounds a disc without punctures, $\int K = \pi + \varphi > 0$; otherwise β is in the core class and bounds with σ_0 an annulus with $\int K = \oint_{\sigma_0} k_g + \pi - \varphi > 0$; both contradict (I).

The chain. Let k be the largest index with γ in the closed disc bounded by σ_k. If $k = N$, let p be a point of γ of largest ρ; there γ touches the circle $\rho = \rho(p)$ from the side of C_{12}, which Lemma 9 with $\tau = -\rho$ and condition C excludes. If $k < N$, some point q of γ is not inside σ_{k+1}. The image of Φ_k is the closed annulus between σ_k and the parallel at time T_k (by injectivity), and it contains every point inside σ_k and not inside σ_{k+1}; on it $t = \Phi_k^{-1}$ is a C^2 function with $|\nabla t| = 1$ ($G > 0$: no focal points) whose level sets are the parallels, convex towards increasing t ($\operatorname{Hess} t(v,v) = G'/G < 0$ on unit tangents). Let p minimise t on the part of γ in the annulus. Then $t(p) \le t(q) < T_k$, so γ stays in the annulus near p and p is a local minimum of $t \circ \gamma$; this contradicts Lemma 9, in the interior if $t(p) > 0$ and at the boundary σ_k (where γ would touch σ_k from inside) if $t(p) = 0$, Φ_k extending to small negative t. □

Stutters. Condition (III) reduces to the normal geodesics of the equator arcs. Let H be an open hemisphere and j an equator arc. Suppose the normal geodesics ν_s of j into H (Fermi coordinates $g = dr^2 + G^2 ds^2$, $G'' = -KG$, $G(0) = 1$, $G'(0) = 0$) form a diffeomorphism of $\{(s, r) : 0 < r < r_{\text{exit}}(s)\}$ onto H, with $G > 0$ and $G' > 0$. Then r is C^2 on H, its level sets are convex towards decreasing r, and Lemma 9 with $\tau = -r$ excludes an interior maximum of r on a geodesic arc in H, hence any stutter arc on j. Independently, Gauss-Bonnet limits the stutters that could occur: when the hemisphere's total positive curvature $\Pi_H = \int_H K^+ \, dA$ is below π there is no geodesic monogon in H ($\int K = \pi + \varphi$), so a stutter arc is simple and bounds with the segment of j between its endpoints (which contains no collision) a disc B with $\int_B K = \theta_p + \theta_q$; both corner angles are then at most Π_H. For equal masses $K \leq 0$ and this recovers the absence of stutters [8, Prop. 2.2].

Numerical status. At the five triples $(1, 1, 0.57)$, $(1, 1, 0.6)$, $(1, 1, 2)$, $(1, 0.5, 0.45)$, $(1, 0.2, 0.19)$, none of which has a waist at any end, (I) is proved and (II), (III) hold numerically, with σ_0 and the chain taken from a curve-shortening flow started near the cuff.

(I) is proved by interval arithmetic. In the stereographic chart ζ centred at a collision C_{ij} the metric is $g = 2U(1 + |\zeta|^2)^{-2}|d\zeta|^2$, and with $F = |\zeta|^2 U(1 + |\zeta|^2)^{-2} = m_i m_j/(2a_{ij}(1 + |\zeta|^2)) + V|\zeta|^2(1 + |\zeta|^2)^{-2}$, where $r_{ij}^2 = a_{ij}(1 - c_{ij} \cdot w)$ and V is the potential of the other two pairs, F is rational, positive and smooth at $\zeta = 0$, and $K < 0$ if and only if $Q = F\,\Delta F - |\nabla F|^2 > 0$ (as $\log|\zeta|^2$ is harmonic). We enclose Q over boxes with second-order interval jets in outward-rounded IEEE arithmetic (only $+, -, \times, /$ and $\sqrt{}$, all correctly rounded), the masses and all derived constants being intervals. The boxes cover, in the chart of each collision, its Voronoi cell (the points no farther from it than from the other two), with adaptive refinement; a box is set aside when it lies wholly outside the cell or wholly inside the polygon σ_0. At all five triples every remaining box has $Q > 0$ (between $1.7 \cdot 10^4$ and $9.3 \cdot 10^5$ boxes per triple), so $K < 0$ on the whole region outside σ_0, the collisions C_{13}, C_{23} included; with a small loop in place of σ_0 the same procedure fails, as it must. With the masses as wide intervals the same proof covers whole intervals of masses at once: with the single loop σ_0 of $(1, 1, 0.57)$, $K < 0$ outside σ_0 for every mass triple $(1, 1, m_3)$ with $0.567 \leq m_3 \leq 1.07$ (173 mass intervals of width at most about 0.005). On that range the loop is convex for the metric of each mass, encloses C_{12} only and avoids arc 3 (checked numerically at 300 masses), so it is a loop σ_0 as in the conditions; for m_3 above about 1.077 it is no longer convex. The test whether a box lies inside σ_0 uses floating point with a margin. Numerically, the maximum of K outside σ_0, on a 400×800 grid away from caps of radius 0.01 about the collisions, is between $-6.1 \cdot 10^{-4}$ and $-1.4 \cdot 10^{-5}$ (the last value at $(1, 1, 2)$ is attained at the edge of the caps about C_{13} and C_{23}, where $\kappa_2 \rho^2 = -1.3 \cdot 10^{-5}$; the caps $10^{-3} \leq \rho \leq 0.1$ about all three collisions are checked with (III) and have $K < 0$, and below 10^{-3} Lemma 2 applies), and σ_0 avoids arc 3. (II): chains of 261, 153, 43, 35 and 659 loops, polygons of 200 points with positive discrete curvature, the loops saved whenever some point had moved a g-distance 0.01 (0.001 at $(1, 0.2, 0.19)$, where the loops pass near the very curved end 23); for every pair one T_k, all 200 normals keep $G' \leq -2.9 \cdot 10^{-4}$ on $[0, T_k]$, the parallels are simple and nested at 21 times in $[0, T_k]$ (between the samples a pair of double points could only appear and disappear through a tangency of two sheets), the parallel at T_k lies inside the next loop, and condition C holds out to the last loop. A sparser chain fails: the normals from the flat parts of a loop must then run until the slowest normal arrives and lose their convexity, which is why the hypothesis is stated with one time per loop. (III): on each of the three arcs, 160 normals (100 from the arc outside caps of radius 0.05 and 30 from each cap down to $\rho = 10^{-3}$) keep

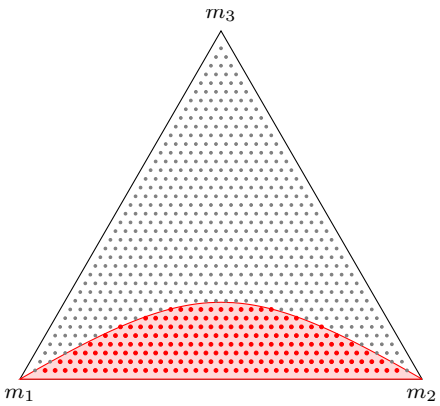


Figure 4: The class (12) over the mass triangle ($m_1 + m_2 + m_3 = 1$). Red dots: a loop shorter than the cuff was found (the class is realised); grey dots: every length minimisation slid into the end. Shaded: the waist region $m_3^2(m_1 + m_2)^3 < m_1^2 m_2^2 M$ of Corollary 4.

$G \geq 1$ and $G' > 0$ and leave the hemisphere in monotone order; the normals from inside a cap stay within 0.1 of their collision, where $K < 0$. The normals that run into the cap of another collision are followed further: each either crosses the equator inside the cap or goes down the end (at $\rho < 0.01$, moving inwards at an angle too steep to turn back by the Clairaut relation of the near-cylinder end), and none returns to the hemisphere; at $(1, 1, 0.57)$, for instance, 10, 10 and 28 normals per arc do so, 3, 3 and 4 of them crossing the equator. Below $\rho = 10^{-3}$ we rely on the near-cylinder form of Lemma 1, without a quantitative estimate. Π_H is between $3.8 \cdot 10^{-4}$ and $1.5 \cdot 10^{-2}$. With a waist, at $(1, 1, 0.5)$, the same check fails on the two arcs ending at C_{12} and passes on the third. The integrations are fixed-step fourth-order Runge-Kutta in arclength, each repeated with half the step, and both step sizes must pass. Conditions (II) and (III) are numerical evidence in double precision; making them rigorous, and covering all masses rather than these triples, is left open. The band just above m^*, where all margins vanish with κ_2, needs a separate argument.

The whole mass triangle. Beyond these triples we mapped the triangle numerically (Figure 4). At 741 mass triples on a grid ($m_i = k_i/40$), we minimised the g-length over loops around C_{12}, star-shaped in the chart about C_{12} with the logarithm of the radius a trigonometric polynomial of degree 12, from three starting circles, with the other two collisions kept outside. By Theorem 6 a loop shorter than the cuff already gives a closed geodesic. Such a loop is found at exactly the 198 triples with a waist at the end 12, where the minimisation stops on the orbit; at the other 543 every minimisation slides into the end with length above the cuff, and no local minimum away from the end appears. The search sees minimisers and local minima only, not closed geodesics of saddle type; the map comes from the research script `tools/words_T0.py` and has no independent checker.

7 Verification

The supplement contains checkers that share no code with the research tools, each ending with ALL PASS:

- the initial conditions of the nine orbits of Table 1 to full double precision, re-integrated with a fixed-step Runge-Kutta scheme (two step sizes) to confirm the invariants, the shape closure, the action and the syzygy word;

- the reversible search of Section 5 at nineteen mass triples, in a regularised time and at two step sizes, with every sign change of the shooting function refined to tell a root from a jump between branches: a root at the start parameter of each of the nine orbits of Table 1, and none at the ten other triples;

- a symbolic check that the curvature criterion is equivalent to the mass condition, gives $8m^2 - m - 2 = 0$ for masses $(1, 1, m)$, and is the limit at the collision of the sign function of [15, (7.13)]; a configuration-space computation of the loop lengths around C_{12} on both sides of the threshold, which also confirms the size of κ_2 through Corollary 4; exact rational expansions of K at a collision;

- for condition (I) at the five triples and on the family $(1, 1, m)$, $0.567 \leq m \leq 1.07$, the interval-arithmetic proof described in Section 6.2, with its negative control;

- for the converse at the five triples of Section 6.2, the chains of convex loops and a checker with its own curvature formula, charts and integrator that re-checks Lemma 2, the absence of a waist at every end, (I), (II) in the form stated, condition C, (III) with the end normals, and Π_H, together with the negative control $(1, 1, 0.5)$.

Two further checkers recompute the traces and rotations of Table 1 (the rotations also as half the enclosed solid angle) and the birth law: $A_4(m^*)$, C, the ratios quoted in Section 5, and $S < \min_\rho \ell$ at all nine masses.

The numerical searches, the symbolic and interval computations and the verification scripts were carried out with the assistance of a large language model. All results are reproducible from the scripts and certificates supplied; the checkers were written independently of the research code, so that they do not share the assumptions of the code that produced the results.

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